

A Project Report On

Founder Link

Submitted in partial fulfillment of the requirement for the
award of the degree

Master of Computer Applications
(MCA)

Academic Year 2026 – 27

Keyur Jada (92500584098)
Punit Nakum (92500584127)
Raj Usadadiya (92500584029)

Internal Guide
Dr Jaypalsinh Gohil



Marwadi
University
Marwadi Chandarana Group





Marwadi
University
Marwadi Chandarana Group



Faculty of Computer Applications (FoCA)

Certificate

This is to certify that the project work entitled
Founder Link
submitted in partial fulfillment of the requirement for
the award of the degree of
Master of Computer Applications (MCA)
of the
Marwadi University
is a result of the bonafide work carried out by

Keyur Jada (92500584098)
Punit Nakum (92500584127)
Raj Usadadiya (92500584029)

during the academic year 2026-27

Faculty Guide

HOD

Dean

DECLARATION

We hereby declare that this project work entitled FounderLink – Startup Team Formation Platform is a record done by us.

We also declare that the matter embodied in this project is genuine work done by us and has not been submitted whether to this University or to any other University / Institute for the fulfillment of the requirement of any course of study.

Place:

Date:

Keyur Jada (92500584098)
Punit Nakum (92500584127)
Raj Usadadiya (92500584029)

Signature: _____
Signature: _____
Signature: _____

ACKNOWLEDGEMENT

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Keyur Jada (92500584098)
Punit Nakum (92500584127)
Raj Usadadiya (92500584029)

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1. Introduction to Project Definition

Startup co-founder matching platform is a full-stack web application developed using the mern stack (mongodb, express.js, react.js and node.js). The platform is designed to connect startup founders with talented individuals like developers, designers, marketers, and business professionals who are interested in joining startup teams.

Many aspiring entrepreneurs have innovative business ideas but struggle to build their startups due to a lack of technical expertise, team members, or business acumen. At the same time, many skilled professionals and students are eager to work on real startup projects but lack business ideas of their own. This platform bridges this gap by providing a centralized environment where founders can post their startup ideas and skilled users can discover these projects and apply to join.

This system provides secure user authentication, startup posting, application management, profile management, and team collaboration features. It allows founders to review applications, accept or reject candidates, and build strong startup teams. This application follows a modern client-server architecture with a restful api, which ensures scalability, security, and high performance.

Using the mern stack, this platform offers a responsive user interface, fast data processing, and efficient database management. It provides an excellent solution for fostering innovation, collaboration, and entrepreneurship, while also giving users valuable real-world experience in startup development.

2. PREAMBLE

Startup co-founder matching platform is a mern stack web application designed to connect startup founders with skilled professionals such as developers, designers, marketers, and business experts. The platform provides an efficient way for founders to build teams by posting startup ideas, while candidates can search for opportunities and apply based on their skills and interests. It provides a secure, user-friendly, and responsive environment that fosters collaboration and innovation. This system simplifies the process of finding the right co-founders and team members, helping to transform startup ideas into successful businesses.

2.1 module description

1. User authentication module

- ✓ User registration and login.
- ✓ Secure authentication using jwt.
- ✓ Password encryption and role-based access control.

2. User profile module

- ✓ Create and update user profiles.
- ✓ Add skills, experience, and contact information.
- ✓ Show profile details to founders.

3. Startup management module

- ✓ Founders can create, edit, and delete startup posts.
- ✓ Specify startup description, required skills, and roles.
- ✓ View all posted startup ideas.

4. Application management module

- ✓ Candidates can apply for startup opportunities.
- ✓ Founders can view applications.
- ✓ Accept or reject candidate requests.

5. Team management module

- ✓ Manage selected team members.
- ✓ Display accepted co-founders and contributors.
- ✓ Track team composition for each startup.

6. Dashboard module

- ✓ Personal dashboard for founders and candidates.
- ✓ View startup posts, applications, and team status.
- ✓ Access important project information in one place.

7. Admin module (optional)

- ✓ Manage users and startup posts.
- ✓ Remove inappropriate content.
- ✓ Monitor overall platform activities.

3. REVIEW OF LITERATURE

Analysis, study and comparison of similar systems

A literature review was conducted to study existing platforms that connect entrepreneurs, professionals and freelancers. The analysis helped identify their strengths, limitations and opportunities for improvement.

1. LinkedIn

Description: a professional networking platform used for job searching, recruiting, and professional networking.

Strengths:

- ✓ Large professional user base.
- ✓ Skill-based profiles and networking.
- ✓ Easy communication between professionals.

Limitations:

- ✓ Focuses primarily on jobs rather than building a startup team.
- ✓ Does not offer dedicated co-founder matching features.

2. Angellist (wellfound)

Description: a platform where startups can hire employees and connect with investors.

Strengths:

- ✓ Startup-focused recruiting.
- ✓ Startup profiles and job postings.
- ✓ Investor networking opportunities.

Limitations:

- ✓ Designed primarily for hiring employees.
- ✓ Limited support for finding co-founders or building startup teams.

3. Upwork

Description: an online freelancing platform that connects clients with freelancers.

Strengths:

- ✓ Large pool of skilled professionals.
- ✓ Secure payment system.

- ✓ Easy project management.

Limitations:

- ✓ Focuses on paid freelance work.
- ✓ Does not support long-term startup partnerships or equity-based collaborations.

4. Cofounderslab

Description: a platform specifically designed to help entrepreneurs find co-founders.

Strengths:

- ✓ Dedicated co-founder matching.
- ✓ Startup networking community.
- ✓ Skills-based recommendations.

Limitations:

- ✓ Limited free features.
- ✓ Small user base compared to larger networking platforms.

4. TECHNICAL DESCRIPTION

4.1 Hardware Requirement

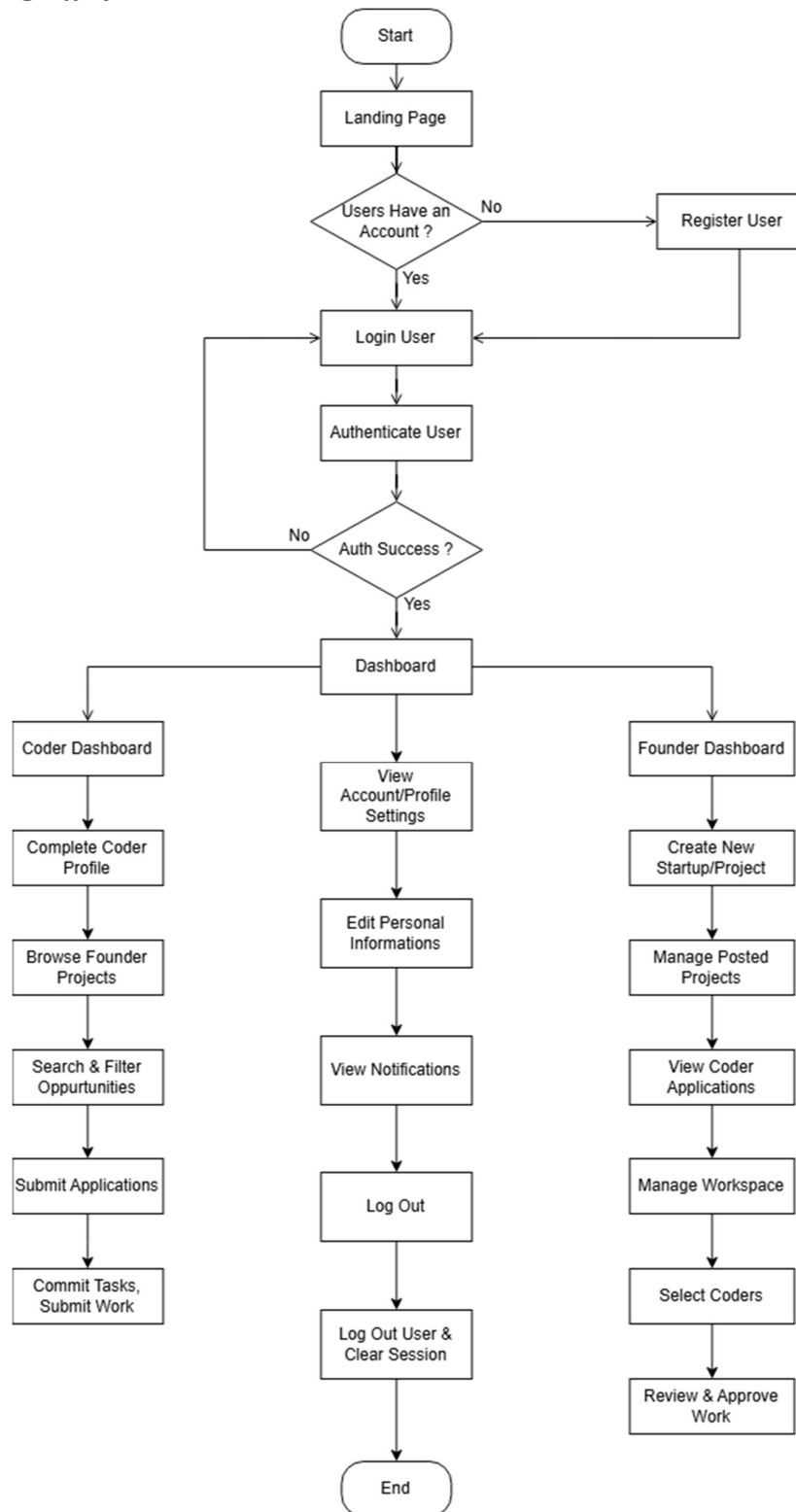
Component	Minimum Requirement
Processor	Intel Core i3 or AMD equivalent
RAM	4 GB (8 GB Recommended)
Hard Disk	20 GB Free Space
Internet Connection	Broadband/Wi-Fi

4.1 Software Requirement

Software	Requirement
Operating System	Windows 10/11, Linux, or macOS
Frontend	React.js, HTML5, CSS3, JavaScript, Bootstrap/Tailwind CSS
Backend	Node.js, Express.js
Database	MongoDB
Runtime Environment	Node.js (v18 or above)
IDE	Visual Studio Code
API Testing Tool	Postman
Version Control	Git and GitHub
Web Browser	Google Chrome, Microsoft Edge, Mozilla Firefox
Package Manager	npm (Node Package Manager)

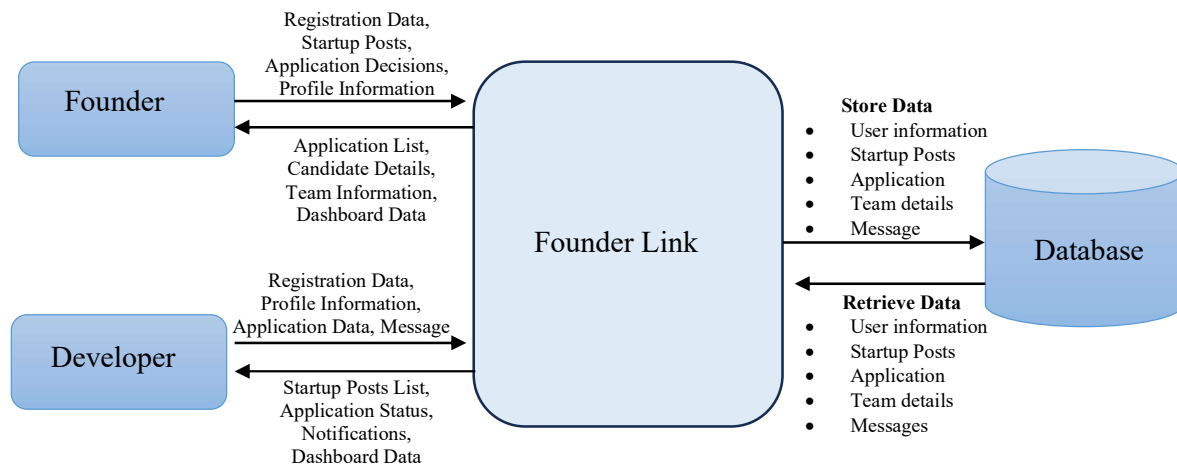
5. System Design and Development

5.1 Flow Chart

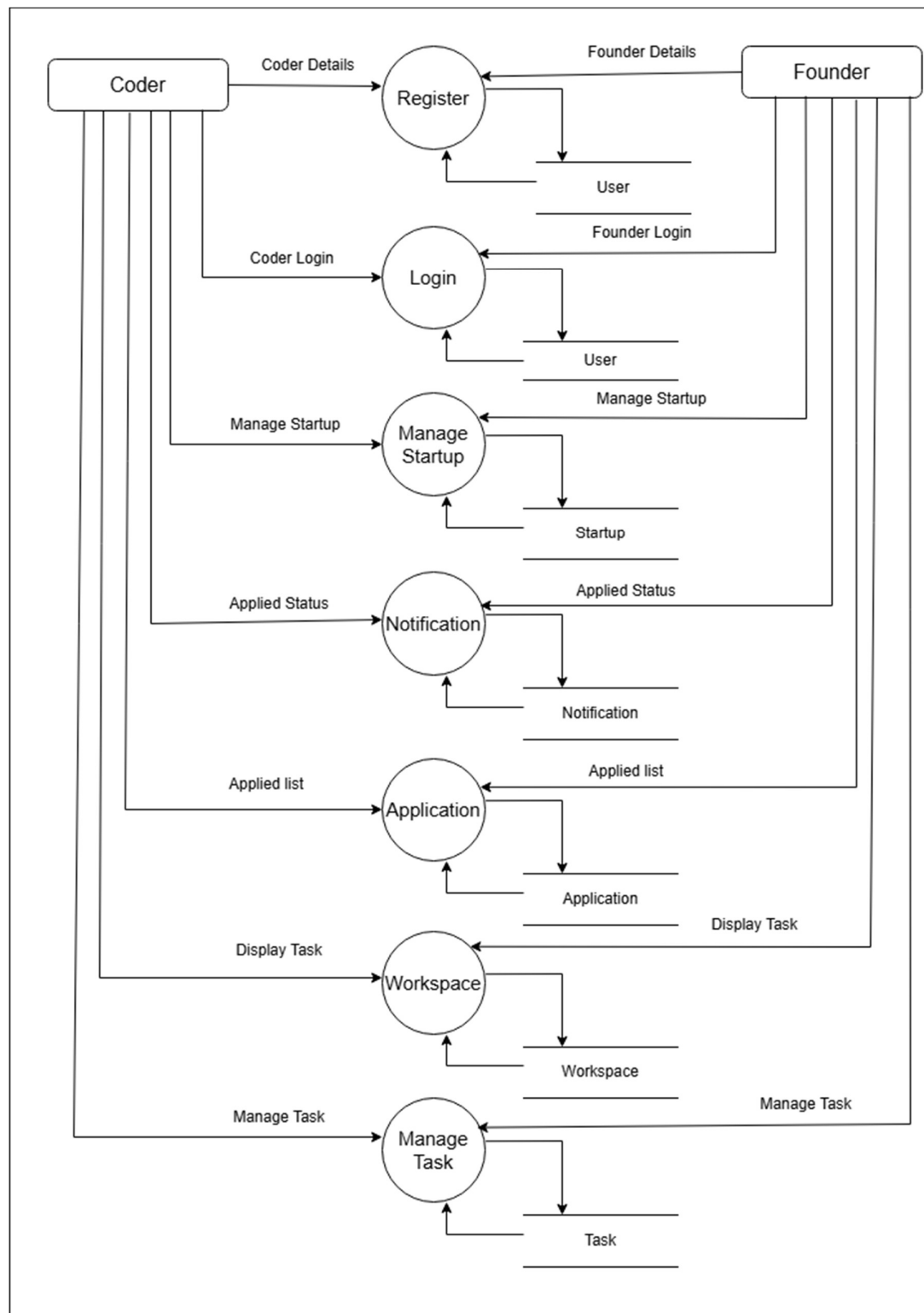


5.2 Data Flow Diagram

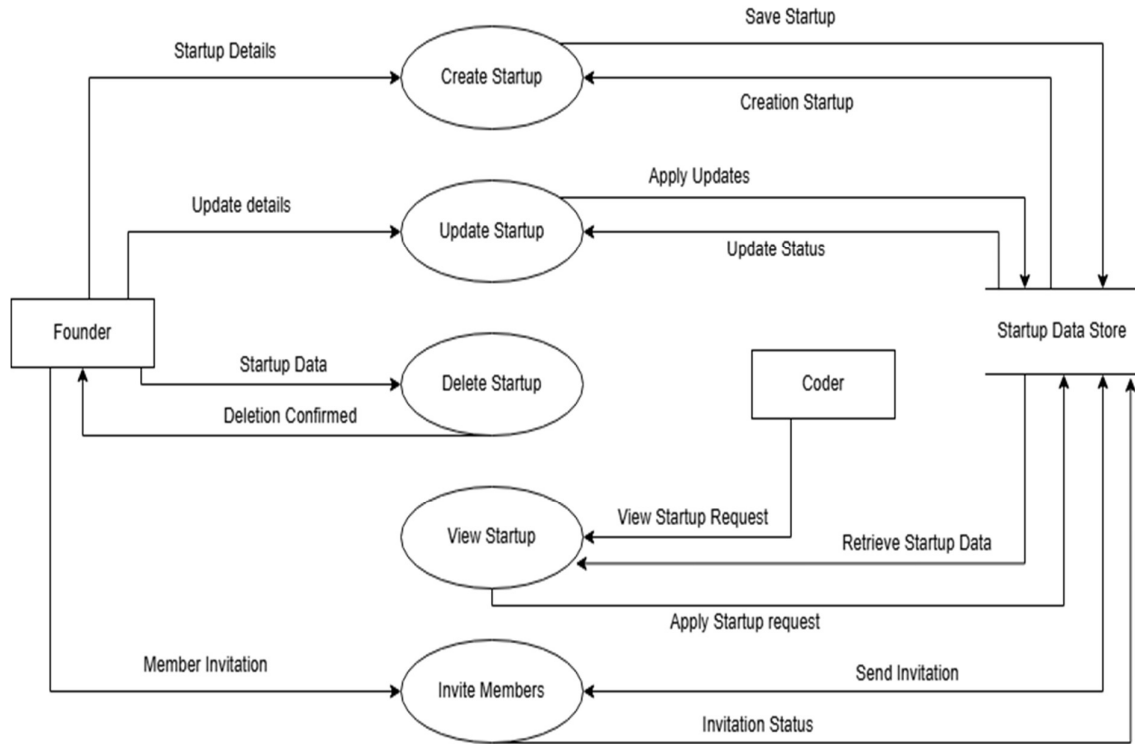
Level 0 DFD: Context Level Data Flow Diagram



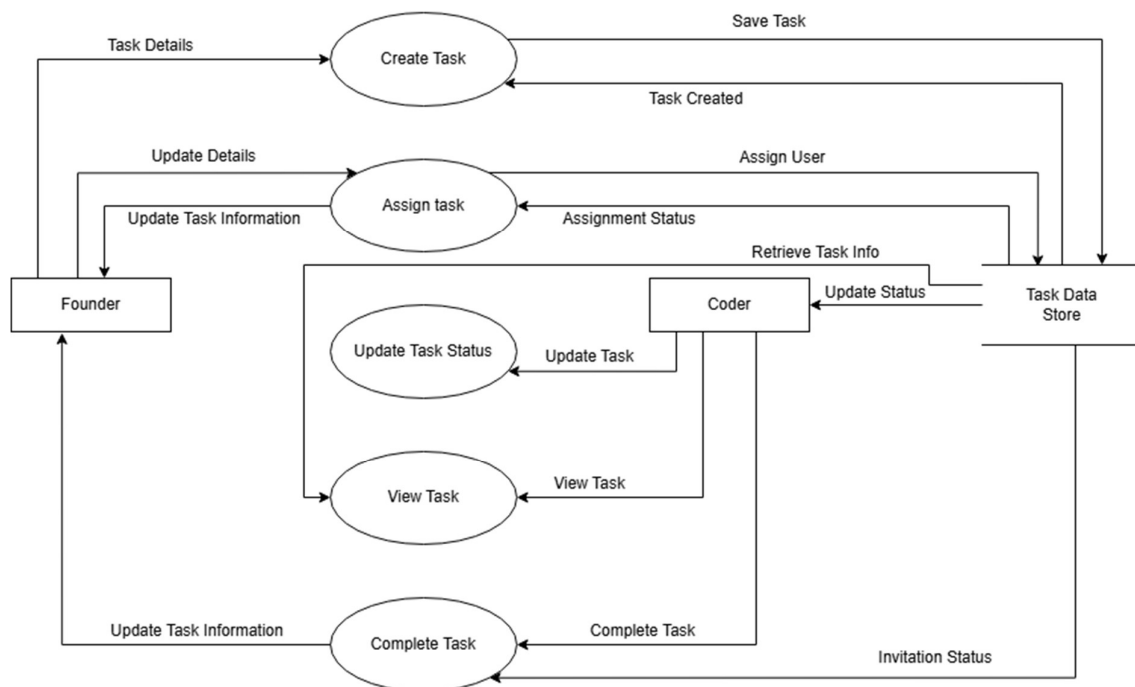
Level 1 DFD: FounderLink System



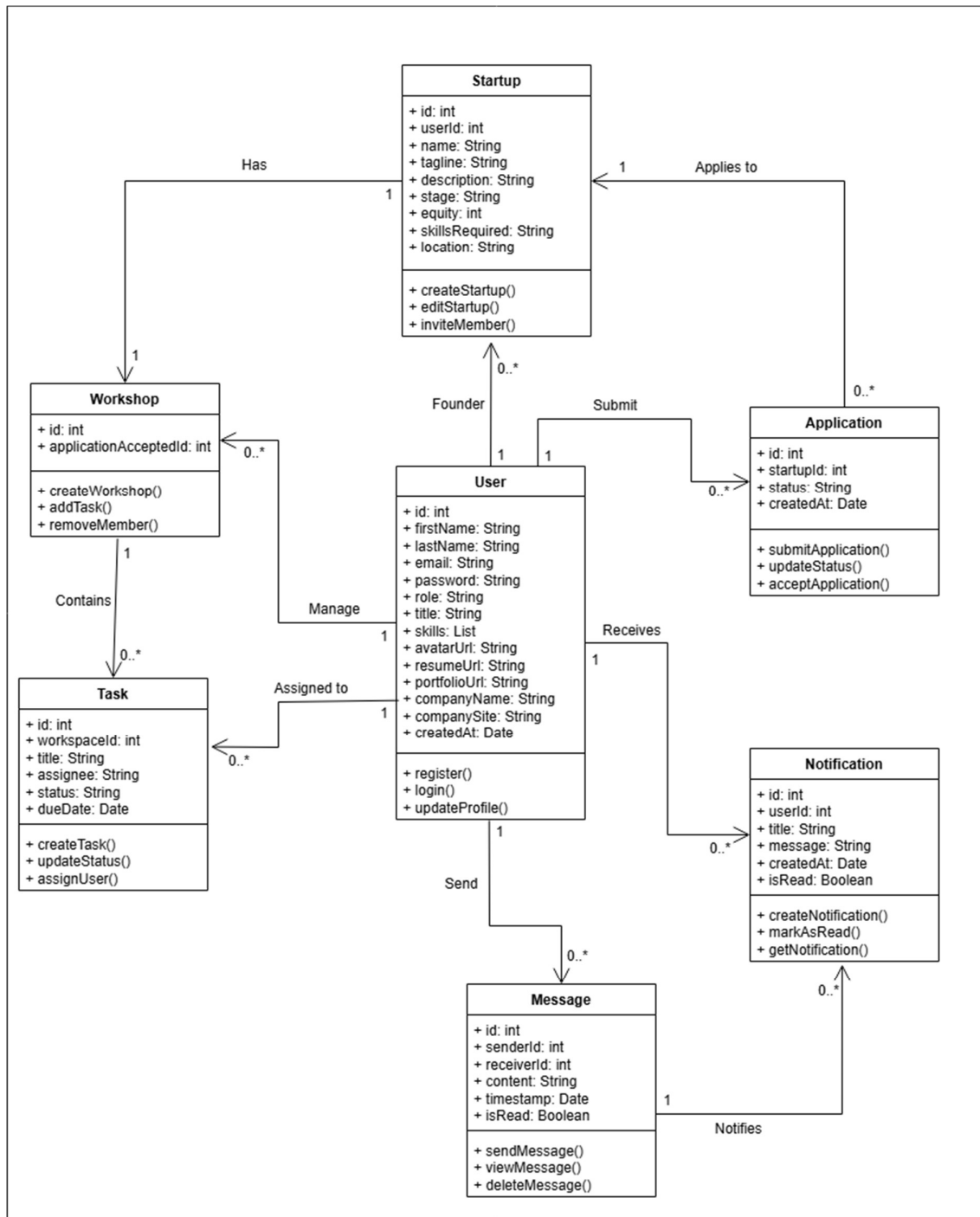
Level 2 DFD: Startup Management



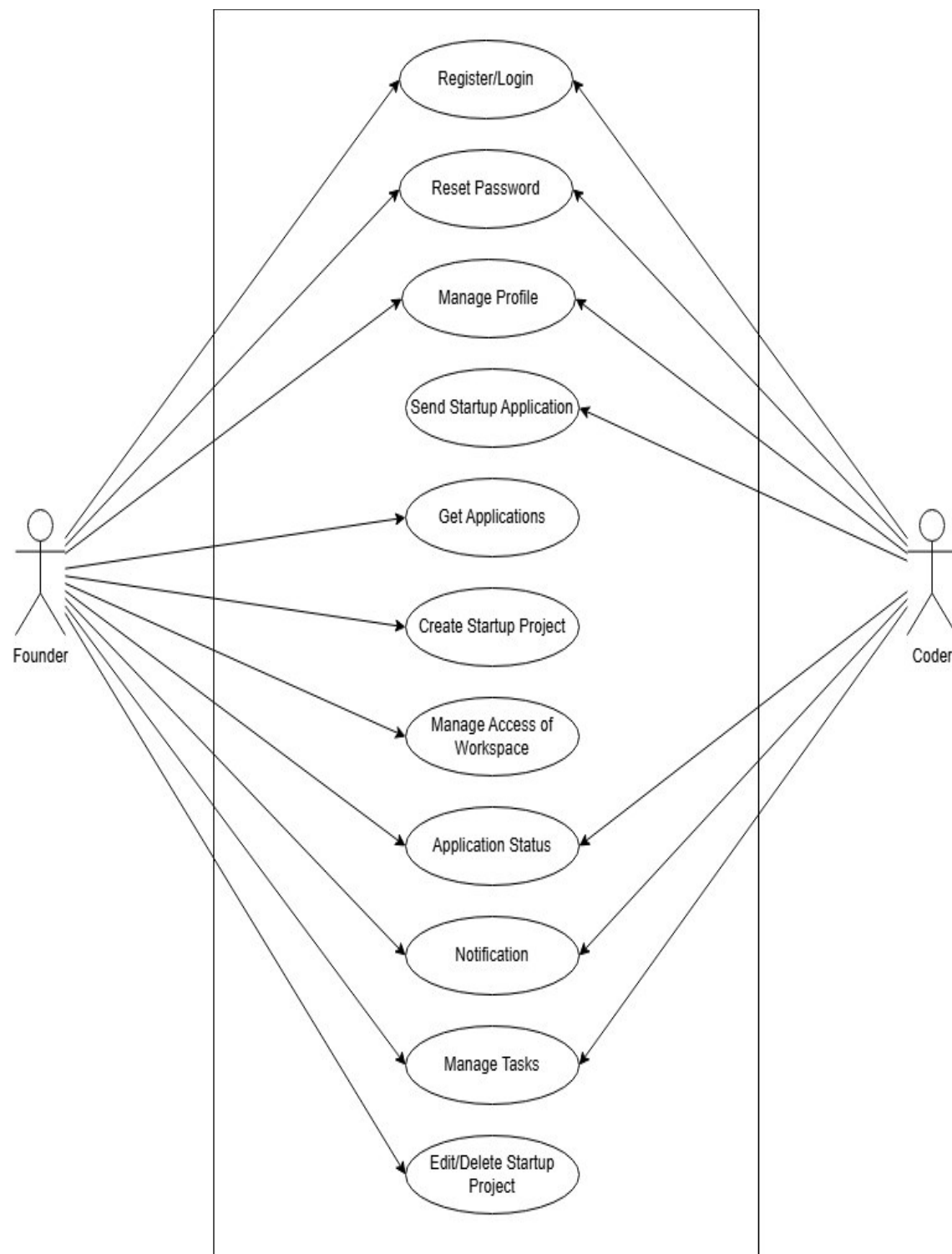
Level 2 DFD: Task Management



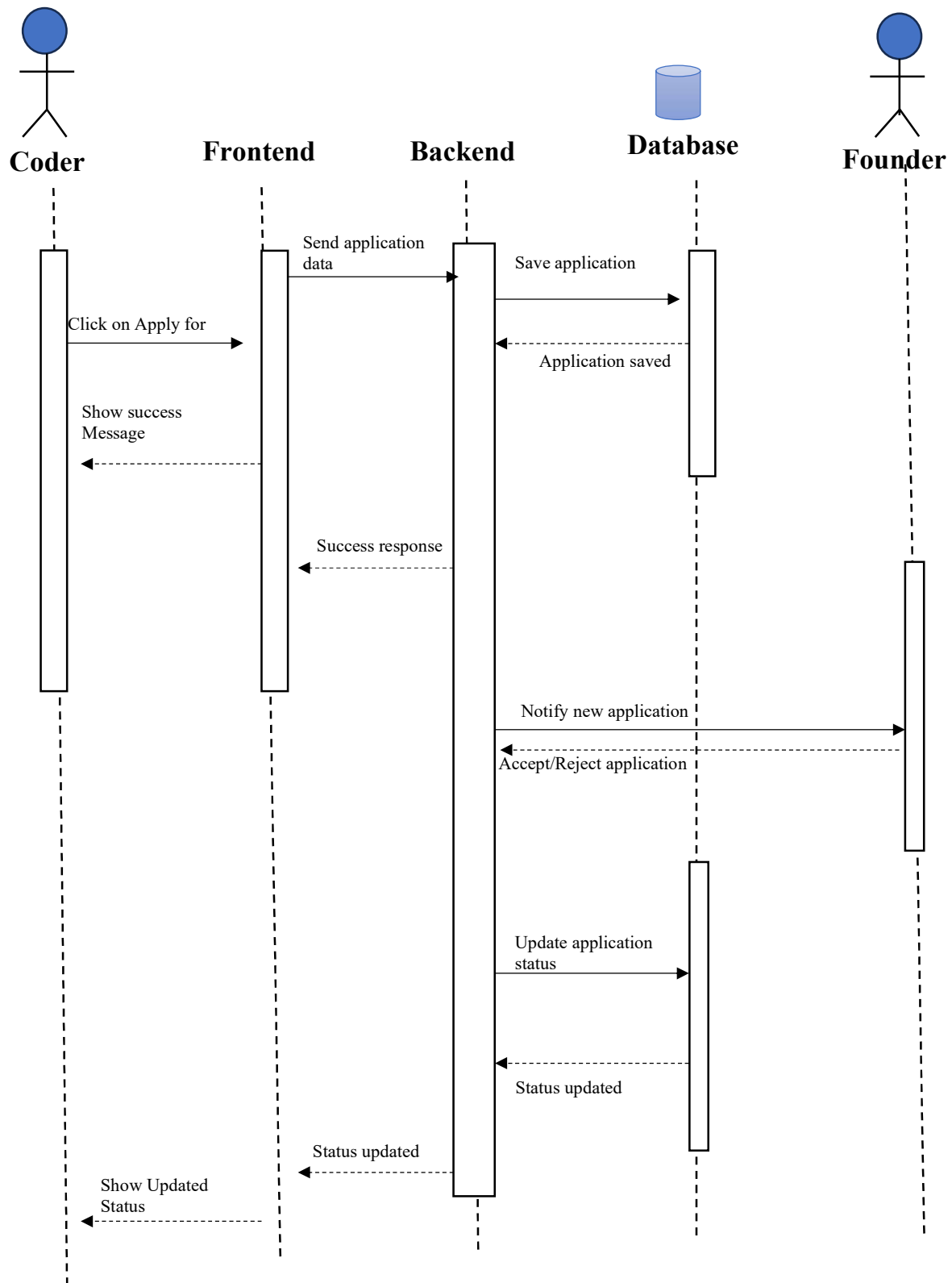
5.3 Class Diagram



5.4 Use Case Diagram

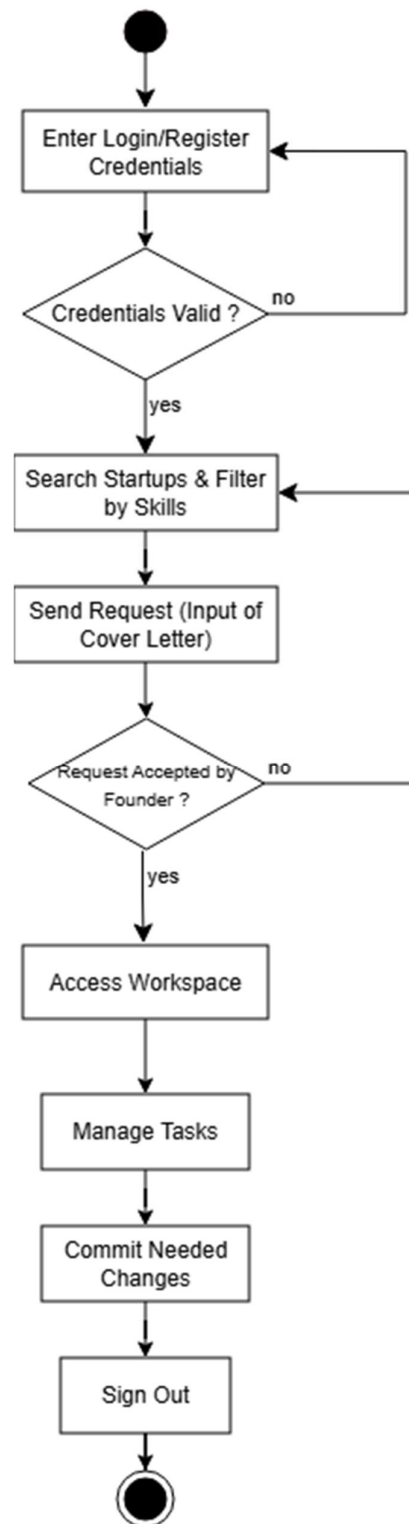


5.5 Sequential Diagram

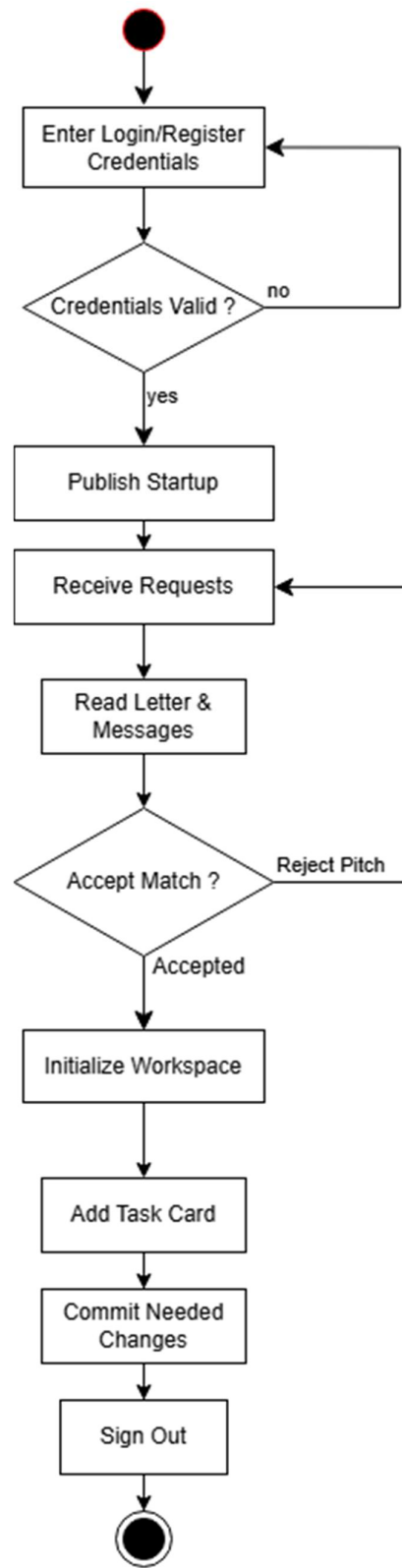


5.6 Activity Diagram

Activity Diagram: Coder



Activity Diagram: Founder



5.7 Database Design / File Structure

Users Table

Field Name	Data Type	Key Constraint	Description
_id	ObjectId	Primary Key	Unique identifier for each user profile
name	String	Required	User's full name
email	String	Unique, Required	User's email address
password	String	Required	Bcrypt hashed password
role	String	Required	Talent / Founder
title	String	Required	Professional headline/job title
skills	Array (String)	Optional	Tech stack and soft skills
avatarUrl	String	Optional	Profile image URL
siteUrl	String	Required	Startup site url
createdAt	Date	System Generated	Account creation timestamp
resumeUrl	String	Required	Resume URL
portfolioUrl	String	Optional	Portfolio's URL
companyName	String	Required	Name of Company

Startups Table

Field Name	Data Type	Key Constraint	Description
_id	ObjectId	Primary Key	Startup ID
name	String	Required	Startup name
tagline	String	Required	Company tagline
description	String	Required	Startup description
stage	String	Required	Startup stage (Seed, MVP Demo, etc.)
equity	String	Required	Equity offered
founderId	ObjectId	Foreign Key (Users)	Founder ID
skillsRequired	Array (String)	Required	Required skills
location	String	Required	Startup location

Applications Table

Field Name	Data Type	Key Constraint	Description
<u>id</u>	ObjectId	Primary Key	Application ID
startupId	ObjectId	Foreign Key (Startups)	Startup ID
applicantId	ObjectId	Foreign Key (Users)	Applicant ID
RequestText	String	Required	Cover letter/pitch
status	String	Required (Default: Pending)	Pending / Matched / Rejected
createdAt	Date	System Generated	Application timestamp

Tasks Table

Field Name	Data Type	Key Constraint	Description
<u>id</u>	ObjectId	System Generated	Task ID
workspaceId	ObjectId	Foreign Key (Workspace)	
title	String	Required	Task title
assignee	String	Required (Default: Unassigned)	Assigned user
status	String	Required (Default: todo)	todo / inprogress / completed
dueDate	Date	Optional	Due date

Notifications Table

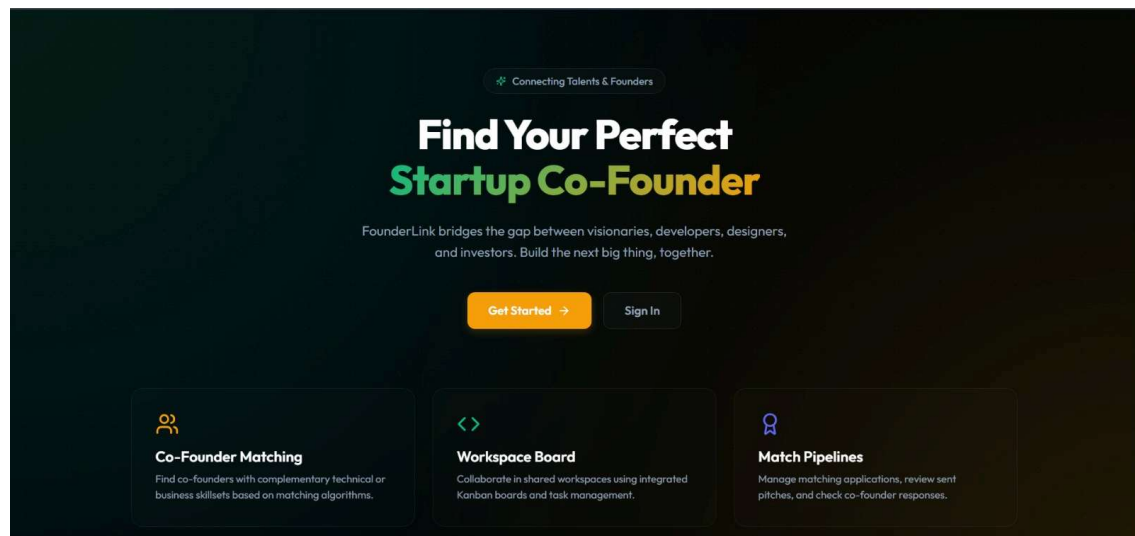
Field Name	Data Type	Key Constraint	Description
<u>id</u>	ObjectId	Primary Key	Notification ID
userId	ObjectId	Foreign Key (Users)	Receiver user ID
title	String	Required	Notification title
desc	String	Required	Notification description
type	String	Required (Default: info)	info / success / warning
time	Date	System Generated	Notification time

Workspace Table

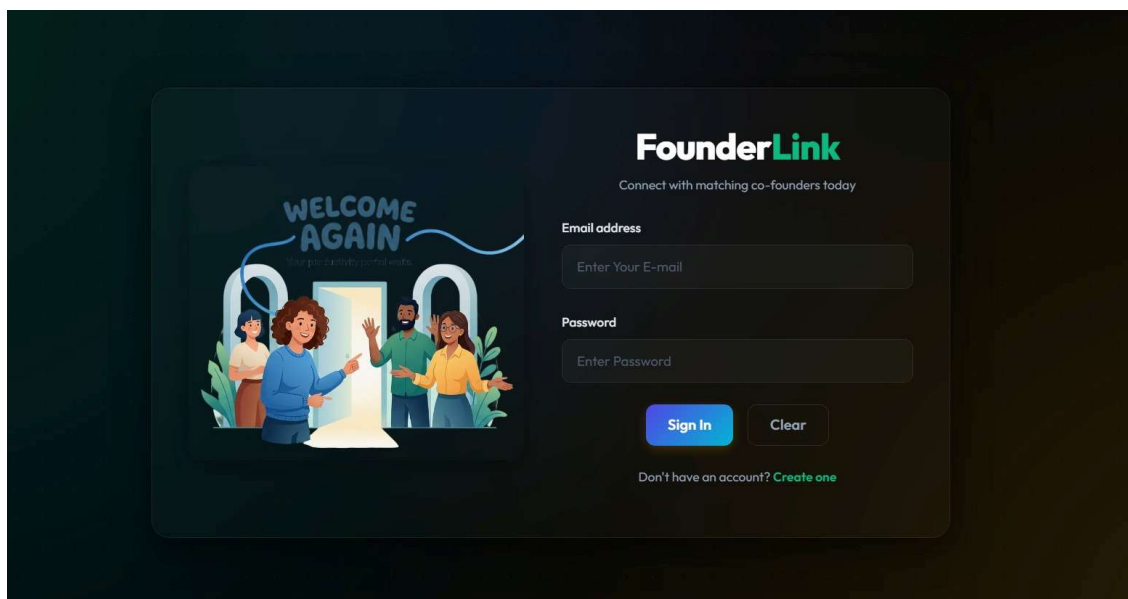
Field Name	Data Type	Key Constraint	Description
<u>id</u>	ObjectId	Primary Key	Team board ID
matchId	ObjectId	Foreign Key (Applications)	Matched application ID
tasks	Array (Task Schema)	Embedded Array	List of task tickets

5.8 Screen Design

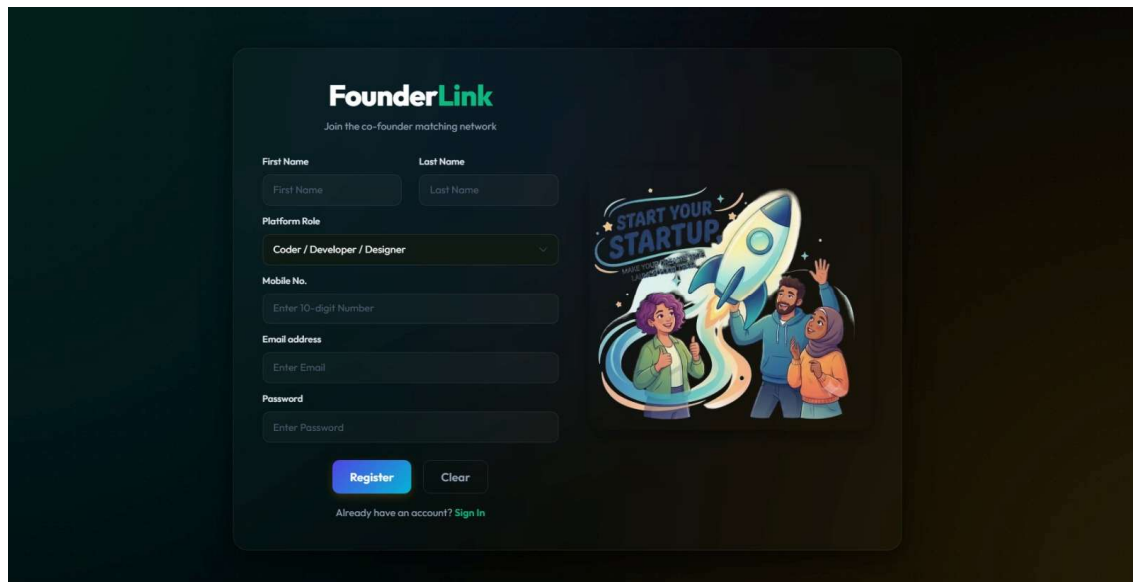
Landing Page



Sign Up Page

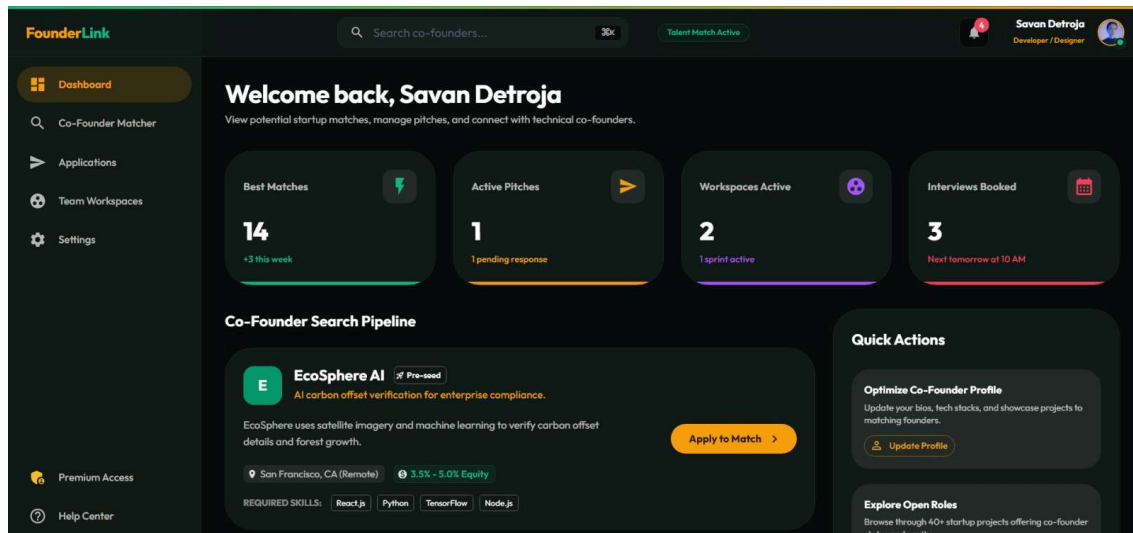


Login Page



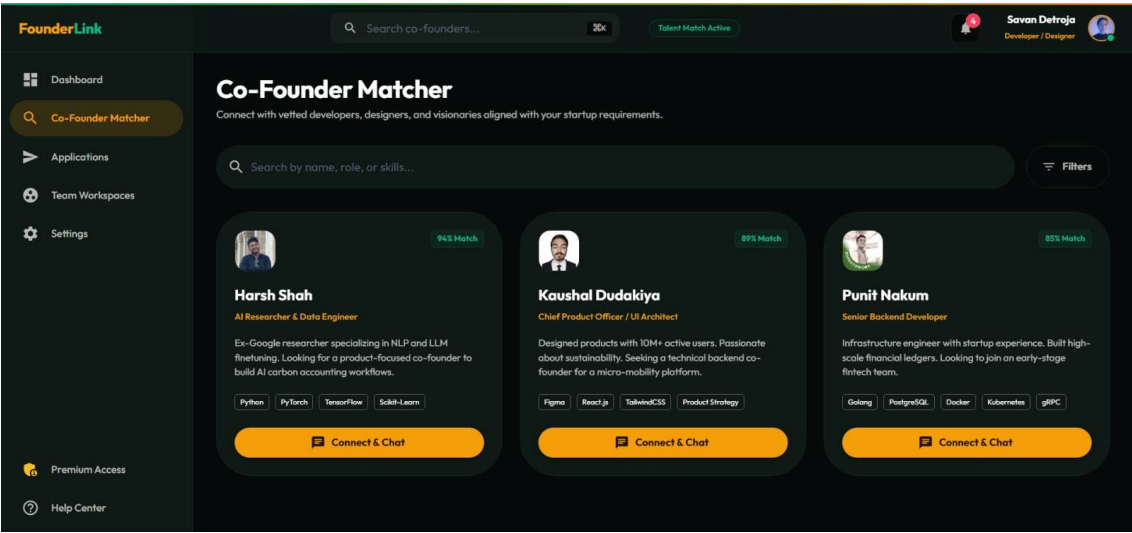
The login page features a dark blue background with a central white form. The form is titled 'FounderLink' in a bold, sans-serif font, with the tagline 'Join the co-founder matching network' below it. The form includes input fields for 'First Name', 'Last Name', 'Platform Role' (a dropdown menu), 'Mobile No.' (with a hint 'Enter 10-digit Number'), 'Email address' (with a hint 'Enter Email'), and 'Password' (with a hint 'Enter Password'). To the right of the form is a colorful illustration of three people (two men and one woman) celebrating the launch of a rocket, with the text 'START YOUR STARTUP' above them. Below the form are two buttons: 'Register' (in blue) and 'Clear' (in white). At the bottom of the form, there is a link: 'Already have an account? [Sign in](#)'.

Dashboard Page

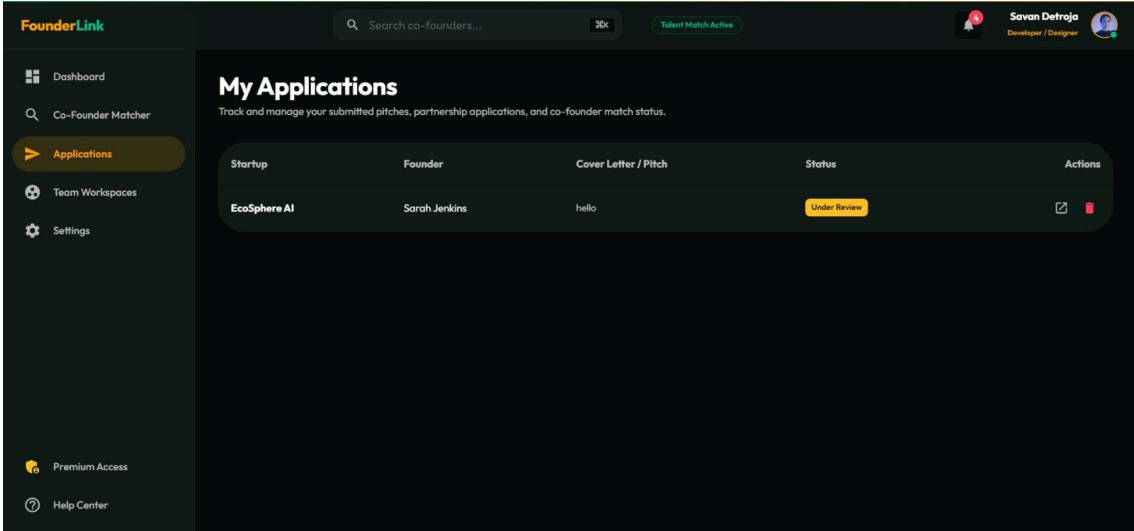


The dashboard page has a dark blue background with a white sidebar on the left. The sidebar contains a search bar and a list of navigation items: 'Dashboard', 'Co-Founder Matcher', 'Applications', 'Team Workspaces', 'Settings', 'Premium Access', and 'Help Center'. The main content area is titled 'Welcome back, Savan Detroja' and includes a subtitle 'View potential startup matches, manage pitches, and connect with technical co-founders.' Below this are four cards: 'Best Matches' (14, +3 this week), 'Active Pitches' (1, 1 pending response), 'Workspaces Active' (2, 1 sprint active), and 'Interviews Booked' (3, Next tomorrow at 10 AM). A 'Co-Founder Search Pipeline' section features a card for 'EcoSphere AI' (AI carbon offset verification for enterprise compliance) with a location 'San Francisco, CA (Remote)' and a salary range '\$ 3.5K - 5.0K Equity'. A 'Quick Actions' section includes 'Optimize Co-Founder Profile' (Update your bio, tech stacks, and showcase projects to matching founders) and 'Explore Open Roles' (Browse through 40+ startup projects offering co-founder status and equity).

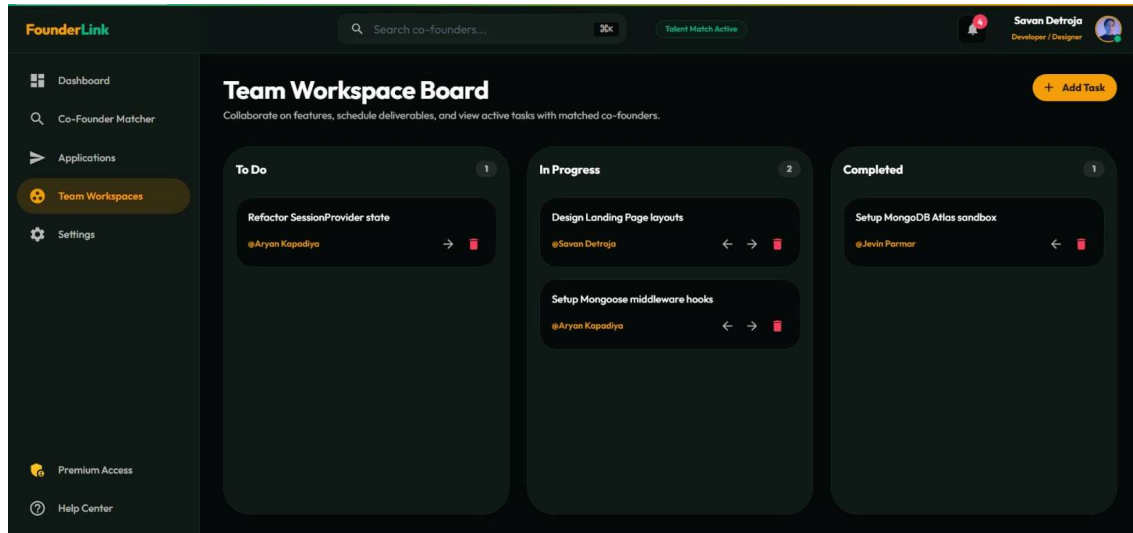
Co-founder match Page



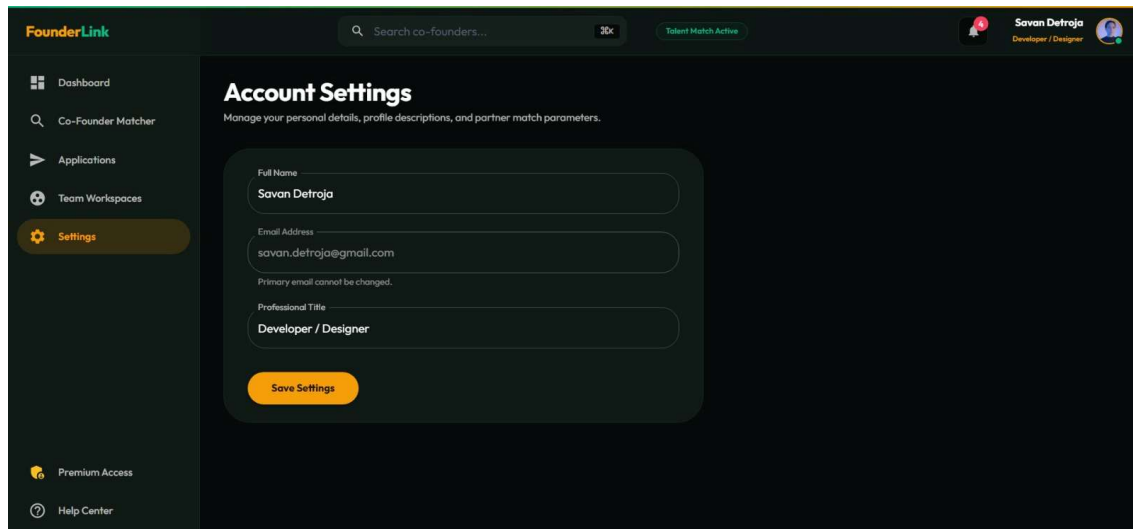
Application Page



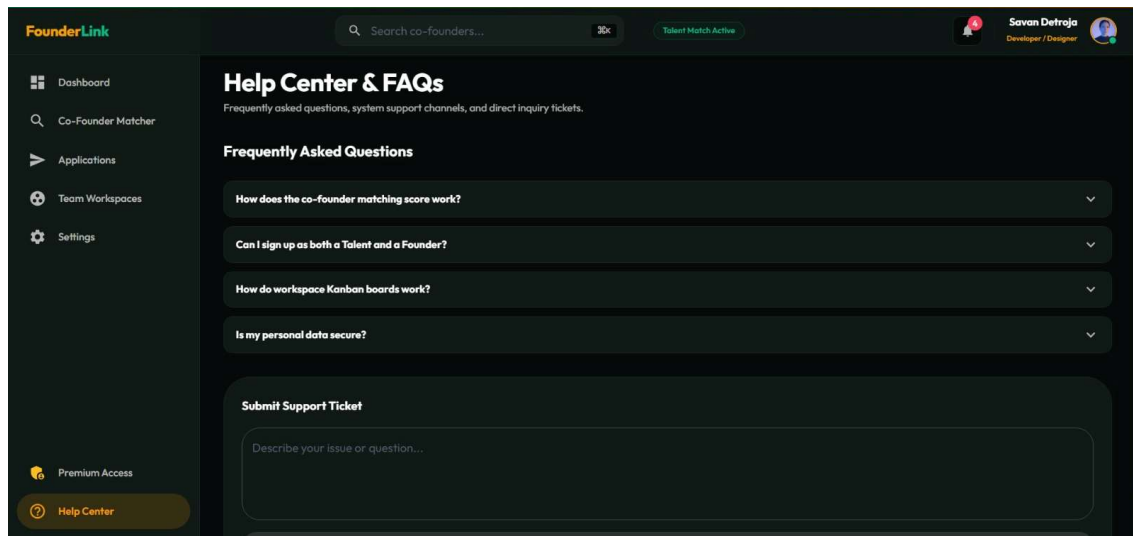
Team Workspace Page



Setting Page



Help center Page



6. SYSTEM TESTING

6.1 Registration & Login Testing

Test case id	Test case objective	Pre requisite	Steps	Input data	Expected output	Actual output	Status
Tc_01	Test user registration	Registration form should be accessible	1. Open registration Page 2. Enter user details 3. Select role4. Submit	Name, email, password, role	User account should be created	User account created successfully	Pass
Tc_02	Test user login	User should be registered	1. Open login page 2. Enter email 3. Enter password 4. Click login	Registered email and password	User should login successfully	User logged in successfully	Pass
Tc_03	Test invalid login	Login page should be accessible	1. Enter email 2. Enter incorrect password 3. Click login	Incorrect email/password	Error message should be displayed	Error message displayed	Pass

6.2 Startup Management Testing

Test case id	Test case objective	Pre requisite	Steps	Input data	Expected output	Actual output	Status
Tc_04	Test create startup	Founder should be logged in	1. Open create startup 2. Enter details 3. Submit	Startup name, description, skills	Startup should be created	Startup created successfully	Pass
Tc_05	Test view startup	Startup should be available	1. Login as coder 2. Open startup section 3. Select startup	Startup name	Startup details should be displayed	Details displayed successfully	Pass
Tc_06	Test update startup	Startup should exist	1. Open startup 2. Edit details 3. Save	Updated startup details	Startup details should be updated	Details updated successfully	Pass
Tc_07	Test delete startup	Startup should exist	1. Select startup 2. Click delete 3. Confirm deletion	Selected startup	Startup should be removed	Startup removed successfully	Pass

6.3 Application Testing

Test case id	Test case objective	Pre requisite	Steps	Input data	Expected output	Actual output	Status
Tc_08	Test startup application	Coder should be logged in	1. Select startup 2. Click apply 3. Enter details 4. Submit	Startup and application details	Application should be submitted	Application submitted successfully	Pass
Tc_09	Test view applications	Founder should have applications	1. Login as founder 2. Open applications	Application details	Applications should be displayed	Applications displayed successfully	Pass
Tc_10	Test accept application	Application should exist	1. Select application 2. Click accept	Application id	Status should change to accepted	Status changed to accepted	Pass
Tc_11	Test reject application	Application should exist	1. Select application 2. Click reject	Application id	Status should change to rejected	Status changed to rejected	Pass

6.4 Notification & Workspace Testing

Test case id	Test case objective	Pre requisite	Steps	Input data	Expected output	Actual output	Status
Tc_12	Test notification	Application status should be changed	1. Change application Status 2. Open notifications	Accepted/rejected status	Appropriate notification should be displayed	Notification displayed successfully	Pass
Tc_13	Test workspace	Coder should be accepted	1. Login as coder 2. Open workspace	Accepted coder	Workspace should be accessible	Workspace accessed successfully	Pass
Tc_14	Test profile update	User should be logged in	1. Open profile 2. Edit details 3. Save	Updated profile details	Profile should be updated	Profile updated successfully	Pass

6.5 Task Management Testing

Test case id	Test case objective	Pre requisite	Steps	Input data	Expected output	Actual output	Status
Tc_15	Test create task	Founder should be logged in	1. Open workspace 2. Create task 3. Assign task	Task details, coder	Task should be created and assigned	Task created successfully	Pass
Tc_16	Test update task	Task should be assigned	1. Login as coder 2. Open tasks 3. Update status	In progress	Task status should be updated	Status updated successfully	Pass
Tc_17	Test complete task	Task should be in progress	1. Open task 2. Complete task 3. Update status	Completed	Task should be marked completed	Task marked completed	Pass

6.6 Database & Logout Testing

Test case id	Test case objective	Pre requisite	Steps	Input data	Expected output	Actual output	Status
Tc_18	Test database operations	Database should be connected	1. Create data 2. Store data 3. Retrieve data	User, startup, application data	Data should be stored and retrieved correctly	Data retrieved correctly	Pass
Tc_19	Test logout	User should be logged in	1. Click logout 2. Try to access dashboard	Logout request	User should be logged out	User logged out successfully	Pass

7. CONCLUSION

Founder link startup co-founder matching platform successfully provides a common platform to connect startup founders with skilled candidates like developers and other team members. The system allows users to register, log in, manage profiles, create and view startup opportunities, submit applications, and manage application status.

The platform also provides notifications, workspace access, and task management, which helps founders and selected team members collaborate effectively. The MERN stack (mongodb, express.js, react.js, and node.js) provides a flexible and efficient environment for developing applications.

Overall, this project helps simplify the process of finding the right startup team members and managing startup collaboration in an organized and user-friendly manner.

8. LEARNING DURING PROJECT WORK

During the development of Founder Link - Startup Co-Founder Matching Platform, I gained practical knowledge of full-stack web development and learned how different technologies work together to create complete applications.

- Learned practical implementation of MERN stack.
- Improved knowledge of React.js for developing user interfaces.
- Learned to build REST APIs using Node.js and Express.js.
- Gained practical experience with MongoDB for storing and managing data.
- Learned about user authentication and authorization.
- Improved understanding of frontend and backend integration.
- Learned how to do testing and debugging.
- Gained experience in designing and implementing various project modules.
- Improved problem-solving and coding skills.
- Learned how to manage project development using Git/GitHub.

8.1 Future Enhancement

In the future, the Founder Link system can be improved in the following ways:

- Co-founder Matching: Help founders find suitable members based on skills and needs.
- Online Interview: Provide the option to conduct interviews within the platform.
- Team Discussion: Add group discussion features for startup teams.
- Project Progress: Show the progress of startup activities and tasks.
- Skill-Based Search: Allow founders to search for candidates according to specific skills.
- User Verification: Add verification of user profiles and professional information.
- Project Sharing: Allow founders to easily share project ideas and requirements.
- Feedback System: Provide feedback between founders and team members.
- Mobile Support: Make the platform fully optimized for mobile devices.
- Scalability: Improve the system to support a large number of users and startup projects.

9. BIBLIOGRAPHY

Online References

gemini

<https://gemini.google.com/>

grok

<https://grok.com>

geeks for geeks

<https://www.geeksforgeeks.org/>

Youtube wscubetech

<https://www.youtube.com/@wscubetech>